

AI-Assisted Federal Survey Harmonization

Research Fact Sheet

2026-02-10

The federal government already collects a massive mosaic of survey data. AI can map how those pieces fit together, unlocking new analytical power without collecting a single additional data point.

What’s the problem?

The federal statistical system operates **48+ surveys** containing over **7,000 overlapping questions**, designed independently by different agencies for different purposes. No complete map exists showing where surveys overlap, where they’re compatible, and where the differences matter. Building that map manually — collecting all instruments, systematically comparing question pairs, classifying harmonization barriers, validating results, and documenting findings — represents months of effort for a skilled research team. This research accomplished that work in weeks using AI-assisted methods, at a total API cost under \$100.

Why should leadership care?

Response rates are declining across federal surveys. Every year, it costs more to collect less. Meanwhile, the government already *has* enormous data that could answer questions no single survey can — if the connections between surveys were mapped and quality-scored.

The primary value is **data enrichment**: using overlapping questions as bridge variables to link surveys into richer analytical datasets. A secondary value is identifying where **instrument consolidation** could reduce respondent burden. Enrichment gives every agency *more* capability. Nobody loses anything.

How is this different?

Traditional survey harmonization is manual, slow, and limited to 2–3 surveys at a time. This research uses **multiple large language models independently rating question pairs**, with ensemble methods and multi-model arbitration to ensure reliability. The approach holds the *entire federal survey topology* in view simultaneously, finding connections no individual expert could surface — not because experts lack intelligence, but because no human holds 7,000 questions in working memory at once.

What does the evidence show?

Validation Check	Result	Interpretation
Do AI raters agree with each other?	3 independent models, substantial agreement	Task is well-defined; results reflect real question properties
Do AI arbitrators converge?	Near-perfect agreement on verdicts ($\kappa = 0.84\text{--}0.90$)	High confidence in final harmonization judgments
Are rates consistent across surveys?	CPS→ACS: 42.5%; FoodAPS→ACS: 48.6%	Stable rates suggest a general property, not a survey-specific artifact
Do easy questions score easy?	Race, age, relationship: near-100% consolidability	Framework has face validity
Does the framework explain <i>why</i> things fail?	Construct mismatch dominates failures (~96%)	Correctly identifies barrier types, not just pass/fail
Is the work reproducible?	Full pipeline, data, and code are public	Any researcher can verify results

Validation Check	Result	Interpretation
Is it cost-effective?	Under \$100 total API cost for 1,598 pair analysis	Orders of magnitude cheaper than equivalent manual review

Key findings

~**45% of questions** across two major survey pairs (CPS and FoodAPS vs. ACS) have at least one viable harmonization path:

- **Tier 1 — Direct recode** (~16%): Close enough to link with simple recoding. High-confidence bridges.
- **Tier 2 — Statistical adjustment** (~29%): Linkable with documented transformations. Usable bridges with known limitations.
- **Tier 3 — Not linkable** (~55%): Different constructs. The framework correctly identifies these and explains why.

What's at stake?

Without this work, the federal survey ecosystem continues as isolated silos. Declining response rates erode quality with no compensating strategy. Cross-survey analytical opportunities — connections that could inform policy without additional collection — remain invisible. Other national statistical offices are already investing in these capabilities.

What's next?

Report 04 extends the analysis from pairwise comparisons to the full multi-survey network, discovering multi-hop enrichment paths (Survey A → B → C) across all 48 surveys — connections that exceed any individual researcher's working memory.

DRAFT — These results are preliminary and have not been independently validated or peer reviewed. Full methodology, data, and code: github.com/brockwebb/federal-survey-concept-mapper. The views expressed are the author's own and do not necessarily represent the views of the U.S. Census Bureau or the U.S. Department of Commerce.